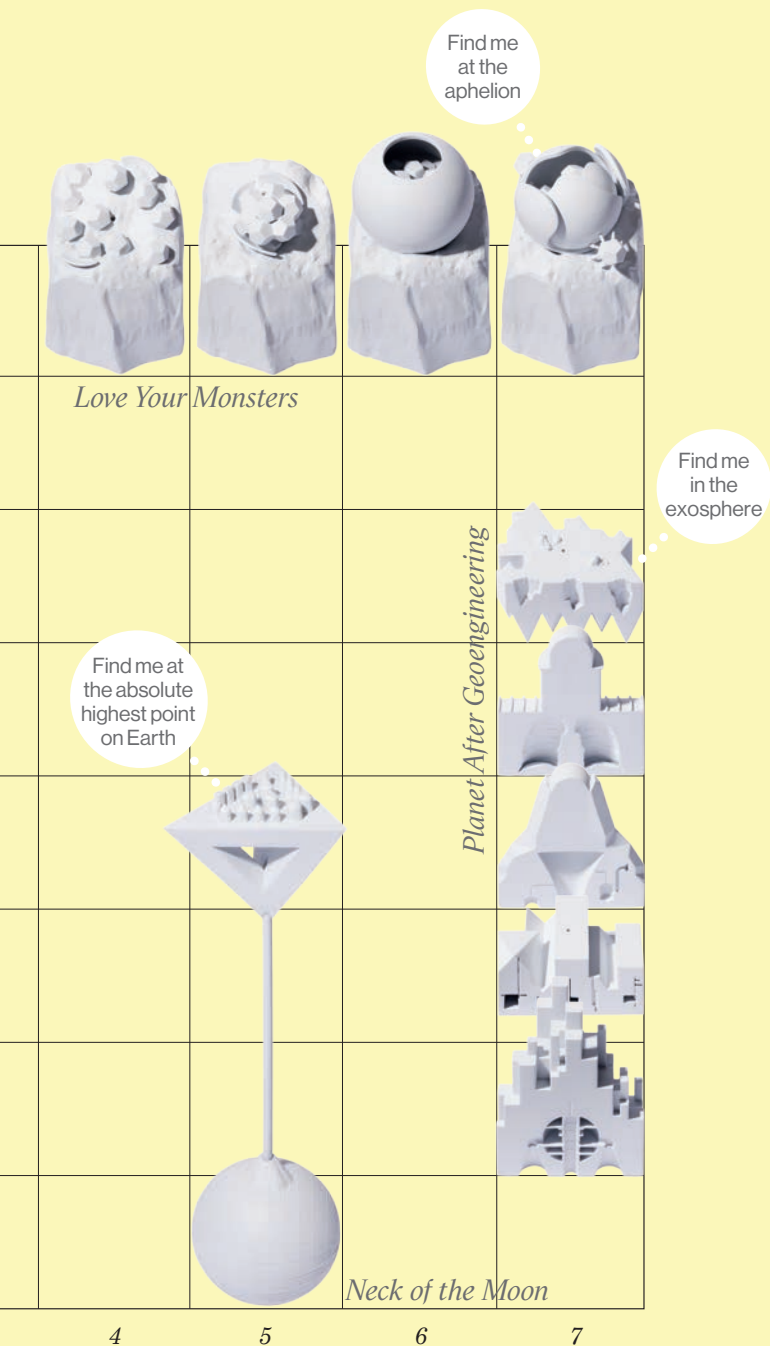
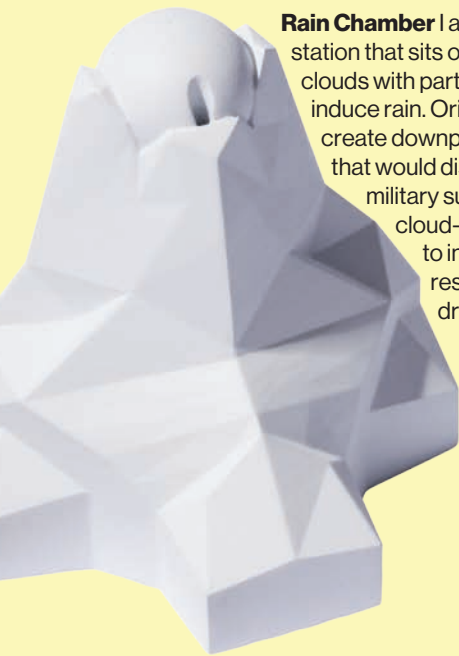


Cosmograph by DESIGN EARTH





Rain Chamber I am a weather modification station that sits on a mountaintop and injects clouds with particles of silver iodide to induce rain. Originally, I was invented to create downpours and cause landslides that would disrupt the North Vietnamese military supply chain. Today, my cloud-seeding technology is used to intercept rain, a precious resource in a time of extreme drought and global warming.



16 Psyche I am an asteroid, domesticated by the 2015 SPACE Act (Spurring Private Aerospace Competitiveness and Entrepreneurship Act). I exist for both material gain—as I am mined for metal ore—and symbolic wealth—as my surface is carved into the faces of the New Space Age gods. Fittingly named after the Greek goddess of the soul, the metal I contain may hold the secrets to the formation of the Earth's core.

Cosmograph
by DESIGN EARTH

MIT
MUSEUM

Cosmograph
Speculative Fictions for
the New Space Age
by DESIGN EARTH

An exhibition organized by the MIT Center for Art,
Science & Technology and the MIT Museum

Cosmograph

Speculative Fictions for the New Space Age

Those who refuse to listen to dragons are probably doomed to spend their lives acting out the nightmares of politicians. We like to think we live in daylight, but half the world is always dark; and fantasy, like poetry, speaks the language of the night.

— Ursula K. Le Guin, *The Language of the Night* (1976)

We are living at the dawn of a New Space Age. Matters of extraterrestrial exploration and exploitation are now the subject of financial speculation, with private companies determined to bring the natural resources of space into the commercial sphere. Outer space—once optimistically considered the province of all humankind—is now being opened up to private enterprise. As extraterrestrial space becomes increasingly within the reach of private gain, new legal, ethical, and spatial questions are arising about its future:

What are we going to do about space junk and extraterrestrial mining?

How can designers relate to the vast scale of a cosmos where Halley's Comet takes 76 years to orbit around the Sun?

What are the environmental implications of the promises of engineering solutions to climate change?

Cosmograph: Speculative Fictions for the New Space Age by DESIGN EARTH explores some of these questions from an artistic point of view. Integrating architectural design, drawing, and science, the artists use the principles of “cosmography” to create speculative narratives for the New Space Age. A term first used by 19th-century explorer and naturalist Alexander von Humboldt, cosmography refers to the combined knowledge from various disciplines, including geography, astronomy, and natural history.

In their four projects, or “geostories” (*Neck of the Moon*, *Love Your Monsters*, *Cosmorama*, and *The Planet After Geoengineering*), DESIGN EARTH combines historical research with storytelling to imagine speculative futures for this emerging era. Using the language of science fiction, these pieces touch upon timely issues like managing space junk, weighing the promises of scientific exploration, navigating geopolitical boundaries, and living with the realities of climate change.

Inviting critical reflection, *Cosmograph* juxtaposes a constellation of artifacts—including large-scale drawings against a backdrop of objects—from the MIT Museum’s collection or on loan. Fictional miniatures are placed strategically to animate the narratives. With a combination of fact and speculation, each narrative brings to light the critical challenges we face—and allows us to envision other possibilities for the future. In these pieces, the vast expanse of outer space provides not only a rich subject for critique, but also a fantasy language through which the artists experiment with new ideas for life on Earth.



About the Artists

DESIGN EARTH is an architectural research practice that deploys the speculative project—drawing and narrative—to foreground the climate crisis as a matter of public concern. Its work has been featured internationally—most recently at Venice Biennale, Bauhaus Museum Dessau, SFMOMA, and Milano Triennale—and is in the New York Museum of Modern Art’s permanent collection. Rania Ghosn and El Hadi Jazairy are the authors of *Geographies of Trash* (2015); *Geostories: Another Architecture for the Environment* (3rd edition, 2022); *The Planet After Geoengineering* (2021); and *Climate Inheritance* (2023).

DESIGN EARTH has been recognized with several awards, including the United States Artists Fellowship, the Architectural League Prize for Young Architects + Designers, the Graham Foundation for Advanced Studies in the Fine Arts Awards, the Association of Collegiate Schools of Architecture Faculty Design Awards for outstanding work in architecture and related environmental design fields as a critical endeavor, and the Graham Foundation for Advanced Studies in the Fine Arts Awards.

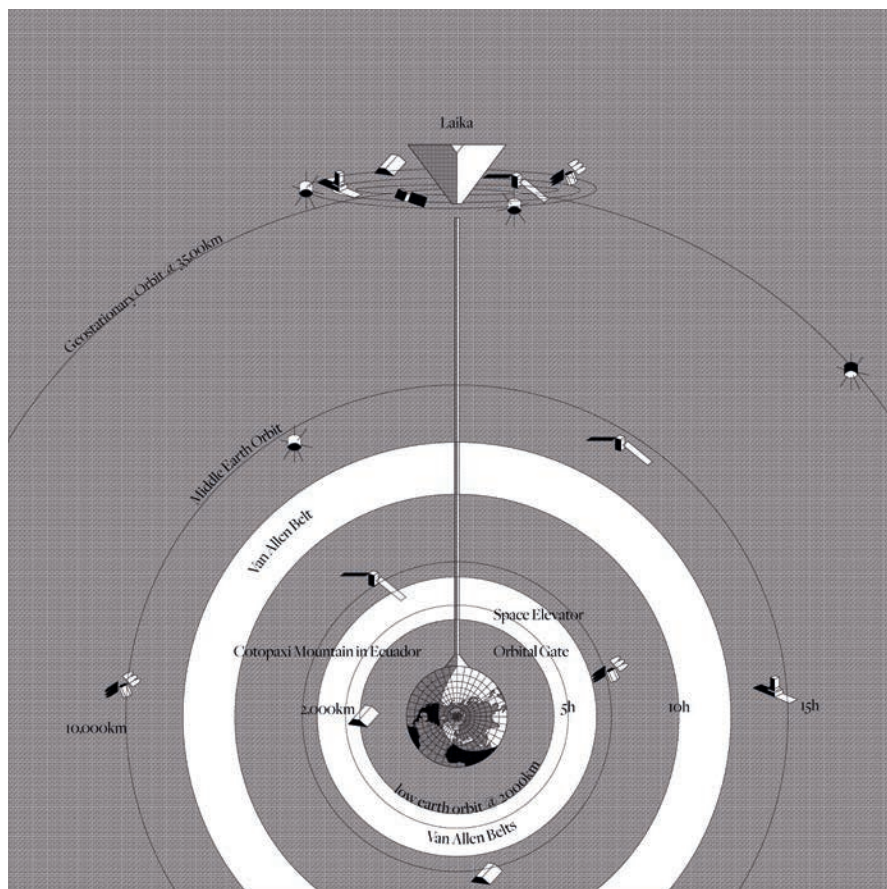
DESIGN EARTH was founded and is directed by Rania Ghosn and El Hadi Jazairy.

Rania Ghosn (b. 1977, Beirut, Lebanon) is Associate Professor of Architecture and Urbanism at the Massachusetts Institute of Technology.

El Hadi Jazairy (b. 1970, Algiers, Algeria) is Associate Professor of Architecture and Director of the Master of Urban Design degree program at the University of Michigan.







Neck of the Moon

2015

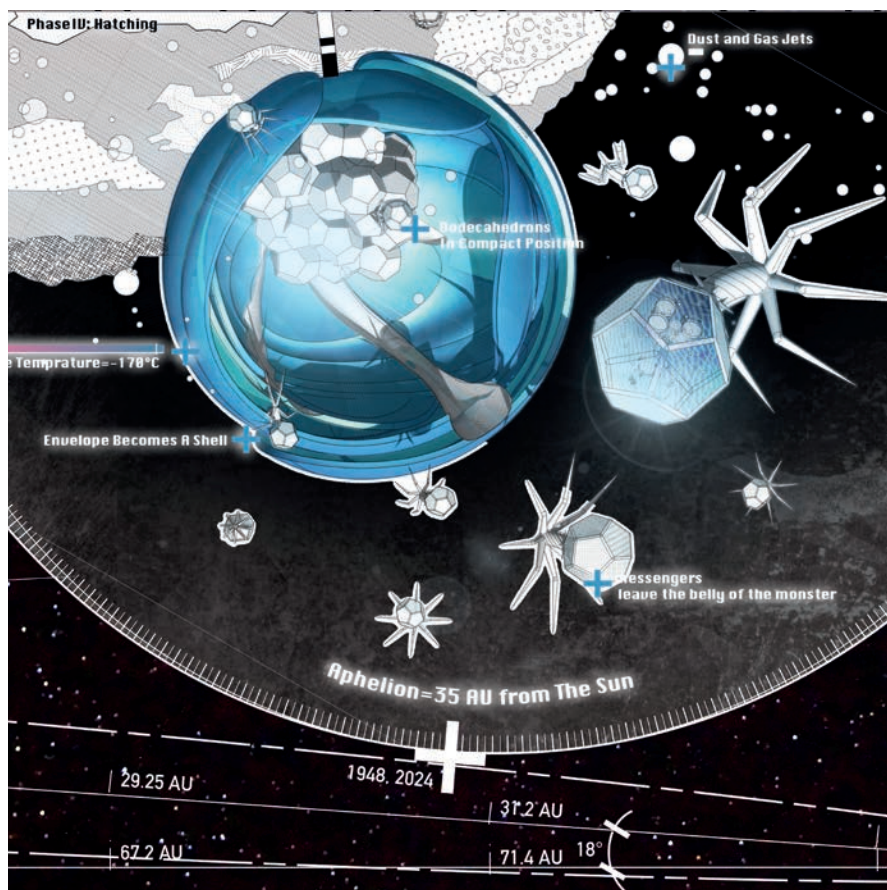
9 drawings (91 x 122 cm), inkjet print on paper

The Space Age began in 1957 with the launch of Sputnik I, the world's first artificial satellite, and its accompanying carrier rocket, which became the world's first piece of space junk. Today, scientists estimate that there are approximately 29,000 large pieces of debris orbiting in outer space. According to both NASA and the European Space Agency, this debris poses a threat not only to other space-based projects but also to people and property on Earth. The number of objects launched into outer space has skyrocketed since 2019. With more than 1,000 new satellites launched into space each year, a number projected to increase to 1,700 by 2030, the concerns associated with orbital debris continue to grow.

Enter Laika, a new satellite planet—named for the stray dog launched into low orbit by the USSR in 1957—made entirely of compacted space junk that orbits the Earth. Equipped with a giant robotic tug that approaches and compresses large pieces of debris at high altitudes, the compacted mass grows steadily and continuously, becoming a kind of second moon. To reduce the cost of launching objects into low orbit, a cord-like structure containing a space elevator tethers Laika to the Earth, allowing oxygen and other materials to be transferred to the new planet, while “stellar energy” generated on Laika’s solar panels is beamed back in return. The *axis mundi* ducks into the deep volcanic crater of Ecuador’s Cotopaxi, one of the highest points on the Earth’s surface. Cotopaxi’s name derives from Quechua, a language indigenous to the equatorial mountains of South America, and means “neck of the moon,” likely in reference to the nights of the year when the moon appears to rest on the volcano’s summit.







Love Your Monsters

2016

5 drawings (61 x 94 cm), inkjet print on paper

In 1531, 1607, and 1682, a “long-haired star” was seen soaring across the night sky, striking existential fear into the hearts of its earthly observers. In 1705, the British astronomer Edmond Halley claimed that all three of these appearances were of the same object—a comet—that orbited the Sun every 76 years. Henceforth called Halley’s Comet, the celestial body would return in 1835, 1910, and 1986. Since ancient times, celestial events like these had been considered signs of impending disaster—the word itself comes from the ancient Greek words *dys*, meaning “bad,” and *astér*, meaning “star.” With the discovery of its regular orbital pattern, however, comets were no longer seen only as omens but also as objects ripe for scientific inquiry.

On its last sojourn, five spacecraft followed Halley’s Comet, capturing thousands of photos of the “dirty snowball” and analyzing the various gases and dust particles that compose it, amassing data that would prove invaluable to future investigations into these celestial objects’ potential for resource extraction.

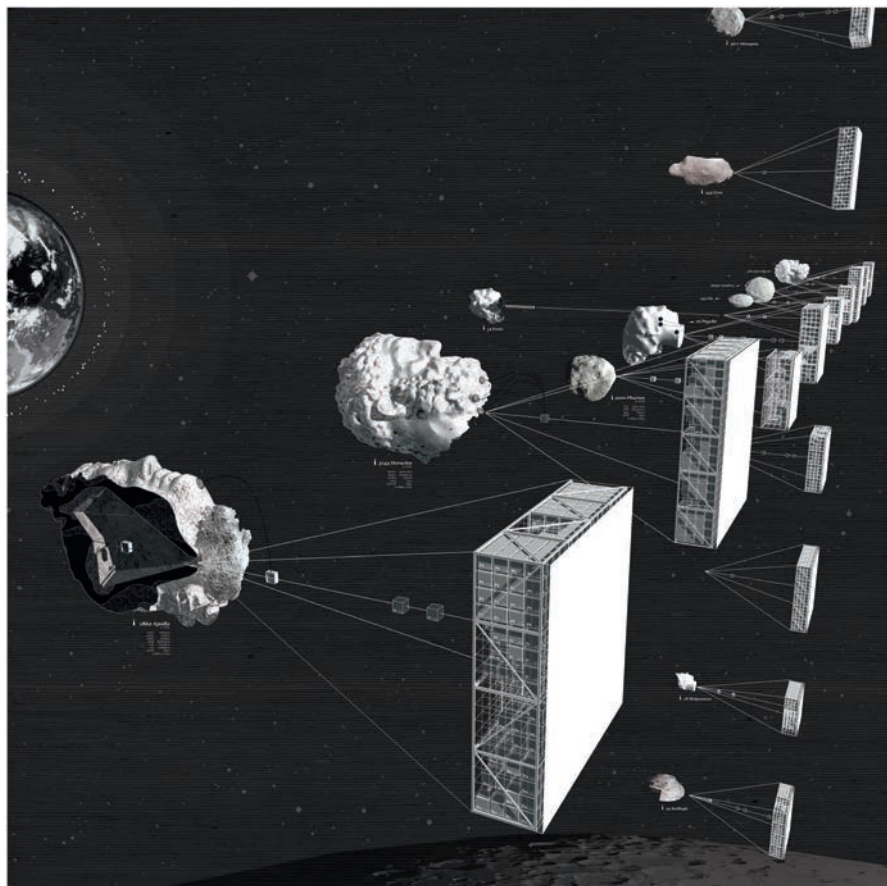
When Halley’s Comet appears in the night sky in 2062, the Halley Armada project will land spacecraft on its surface. Among these spacecraft will be a space probe with a mission patch reading, “Love Your Monsters.” This space probe will be composed of 30 modules, each enveloped in a soft membrane and hosting a living, hairy autonomous reproductive organ.

Landing in the comet’s central depression, the modules will implant a stem to suck water from the comet’s nucleus. As the comet swoops toward the Sun, ice and frozen gases on its sun-facing surface will sublime and the energy released will inflate the membranes of each module, their surfaces harnessing light and heat in an imitation of photosynthesis.

When the comet moves away from the Sun, the envelopes will collapse and the dormant cells will leave their shell and assemble into one large blue egg. While passing through the Oort Cloud at the fringes of the solar system, the egg will collide with trillions of icy celestial bodies and debris and eventually will hatch a swarm of spider-like monsters. This mission will repeat every 76 years until the comet disintegrates into meteor clouds.







Cosmorama

2018

9 drawings (115 x 115 cm), inkjet print on fabric

The passage of the American SPACE Act in 2015 signaled the dawn of a New Space Age in which outer space was no longer “the province of all mankind”—as it was legally defined by the 1967 Outer Space Treaty—but rather the “final frontier” of colonialism and capitalism. As corporate actors create narratives of possible futures for outer space in attempts to solicit capital from investors, the expansion of private industry into the cosmos calls for speculative fictions that complicate and contextualize these futures.

Cosmorama charts this renegotiation of intergalactic territory with three triptychs that imagine new possibilities for political action and assembly on Earth. “Mining the Sky,” “Planetary Ark,” and “Pacific Cemetery” describe, respectively, the concession of asteroid extraction to corporations, the sixth extinction of species due to human activity, and the accumulation of debris in orbit and on Earth. *Cosmorama* combines aspects of astronomy, geography, cartography, and fiction, and shows the relationships among its various components. By drawing together earth and the heavens, these “geostories” posit a correspondence between the politics of the terrestrial and extraterrestrial, giving shape to the ambiguous legal space where the influence of human activity on Earth’s geology and ecosystems has constituted a new geological epoch.







The Planet After Geoengineering

2021

25 drawings (61 x 61 cm), inkjet print on canvas

Graphic animation (13:30 min). Music score by Christine Southworth and Evan Ziporyn.

In the nearly 30 years since the establishment of the United Nations Framework Convention on Climate Change in 1994, the concentration of carbon dioxide in the atmosphere has continued to rise. With the effective mitigation of CO₂ levels becoming increasingly politically inconceivable, the “plan B” offered by geoengineering—or the use of technology to offset the effects of human-induced climate change by intervening in planetary systems—is gaining popularity. However, such large-scale technological proposals remain largely speculative and the long-term consequences of their implementation unproven. Moreover, interventions of this kind are largely irreversible, raising important ethical questions with respect to intergenerational justice and the negative environmental costs.

The Planet After Geoengineering presents a series of fictional narratives depicting Earth—“Petrified Carbon,” “Arctic Albedo,” “Sky River,” “Sulfur Storm,” and “Dust Cloud”—that contextualize geoengineering technologies within a history of climate projects, from 19th-century rainmaking machines and volcanic eruptions to Cold War military plans. Combining three types of media (drawings, animation, and text), *The Planet After Geoengineering* imagines climate-modification technologies as forms of planetary management and idealized global governance. By grounding the abstraction of climate modeling in narrative fiction, these stories make visible the controversies around technological and commercial solutions to climate change, inviting public debate about humanity’s responsibility to all life on Earth.



Museum Collection Objects

1, 2 **Solar Markers** Margaret Benyon, 1979

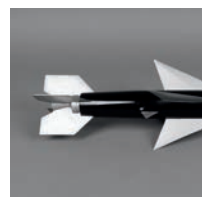
Alluding to the classical elements of earth, air, fire, and water, this mixed media work consists of two burnt rocks (earth and fire) mounted with a hologram showing underwater elements (water) when lit in the open air.

3, 4 **Explorer 11 satellite model** NASA Marshall Space Flight Center, 1962

Explorer 11 was the first gamma-ray satellite with an astronomy telescope used to detect and measure cosmic gamma radiation from space.

5–11 **Rocket models** Various makers, circa 1945–1970

Launching a rocket is difficult. Models like these are important to test qualities on a smaller scale that will be crucial for building full-sized rockets. Some were used to experiment with different rocket propellants, while others were tested in two iterations of MIT's wind tunnel.





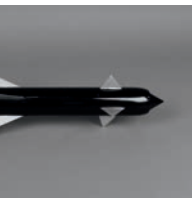
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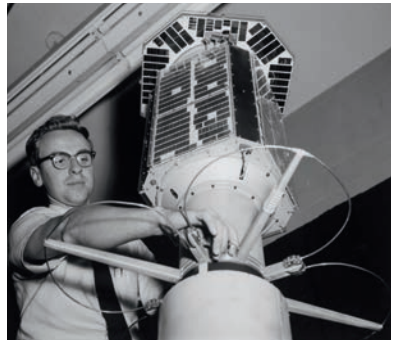
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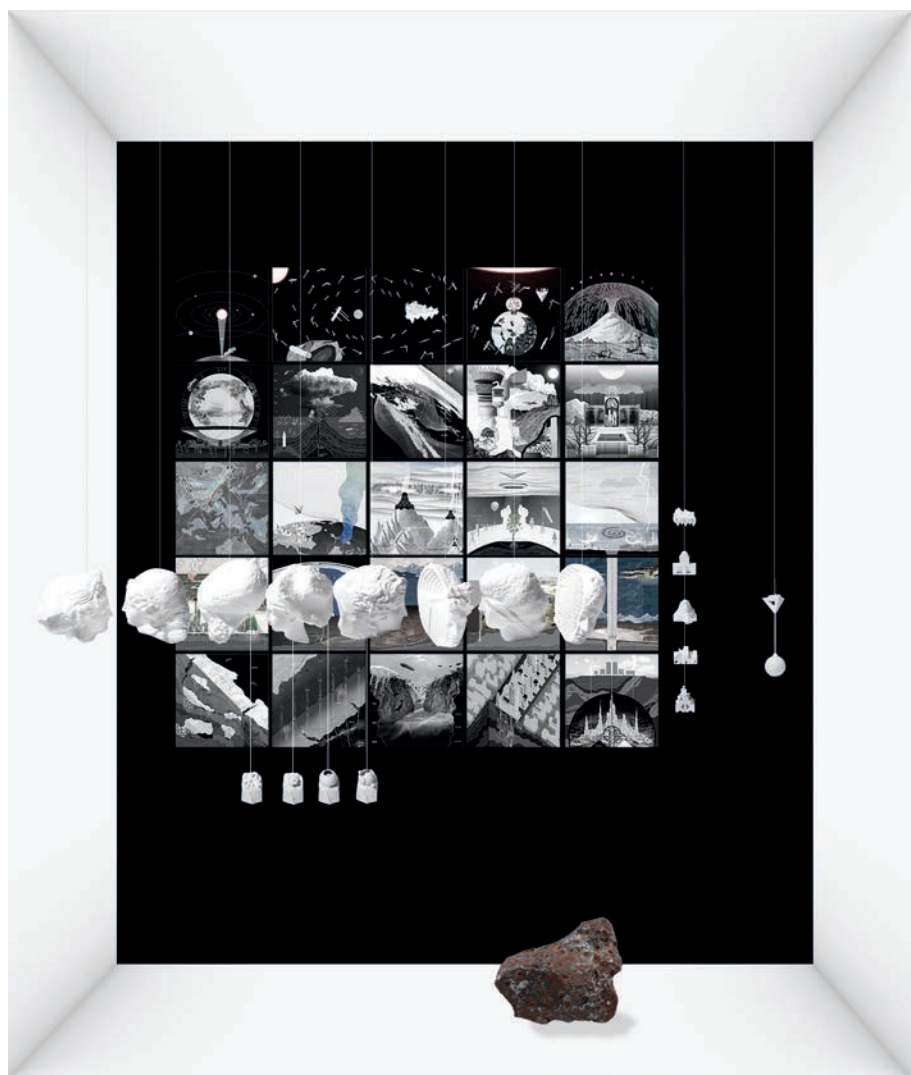


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Acknowledgments from DESIGN EARTH

We are grateful to the DESIGN EARTH collaborators who have worked on the four projects in *Cosmograph*: Rawan Alsaffar, Ayusha Ariana, Garine Boghossian, Reid Fellenbaum, Monica Hutton, Kelly Koh, Meng-Fu Kuo, Hsin-Han (Victor) Lee, Anhong Li, Joude Mabsout, Avery Nguyen, Ranu Singh, Michael Stradley, Ya Suo, Jane Jia Weng, Sihao Xiong, Shuya Xu, Mingchuan Yang, Tianwei Yen, and Bin Zhang.

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For this *Cosmograph* exhibition, we thank Celia Chaussabel (SMArchS '25) for her meticulous overall coordination and design of the 3D objects and this pamphlet, Adam DeTour for the photography of architectural models, and Lily Bartle for her editorial support.

Our gratitude to the CAST and MIT Museum team directors, respectively Leila W. Kinney and Ann Neumann, and to the entire team: Emily Cheeseman, Deborah G. Douglas, Heidi Erickson, Rob Gainfort, Flori Pierri, Leah Talatinian, and Isaac Tardy. Also, to Evan Ziporyn, a wonderful collaborator on *The Planet After Geoengineering* animation, for his role as agent–instigator on this exhibition.

This is first and foremost a work of love by and for companion travelers who seek more just and poetic worlds. To our son, Taha, we love you to the Moon and back!

Credits

The **MIT Center for Art, Science & Technology (CAST)** creates new opportunities for art, science, and technology to thrive as interrelated, mutually informing modes of exploration, knowledge, and discovery. CAST supports visiting artists residencies, art-science research projects, faculty creative work, and curricular initiatives. Evan Ziporyn, Kenan Sahin Distinguished Professor of Music, is the faculty director, and Leila W. Kinney is the executive director.

arts.mit.edu/cast

The **MIT Museum** “turns MIT inside-out,” offering compelling examples of ongoing and past research that raise questions of how science and innovation will shape the future of society. The museum makes innovation available to all by presenting the best of STEAM in its newly designed space in the heart of the Kendall Square Gateway on MIT’s campus. Michael John Gorman is the museum’s director.

mitmuseum.mit.edu

The **CAST x MITM** partnership, launched in September 2023, is an ongoing series of sci/art/tech installations and programs. Programming includes open MIT class sessions and public lectures led by professors and visiting artists on topics ranging from drawing to design and music technology.

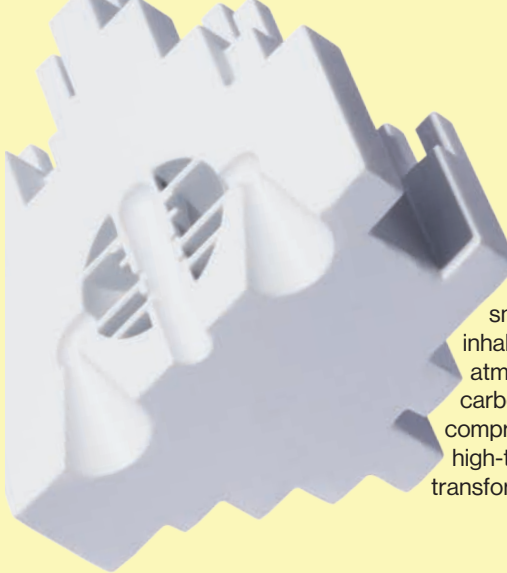
arts.mit.edu/cast-x-mit-museum

Miniature objects photography by Adam DeTour. Courtesy of DESIGN EARTH.

MIT Museum Collection photographs: Explorer 11 satellite model, courtesy MIT Museum. Solar Markers, photo by Margaret Benyon, courtesy MIT Museum. Rocket models, courtesy MIT Museum.

**MIT Museum
314 Main Street
Gambrill Center
Cambridge, MA 02142**

mitmuseum.mit.edu



City of the Captive Carbon Globe

I was built underground as a Cold War nuclear shelter. Now, smog-eating towers sit above me, inhaling excess carbon dioxide from the atmosphere, and I am their lungs. Once carbon enters my chamber, it is compressed in a high-pressure, high-temperature process and transformed into diamonds.



1862 Apollo

I am a “potentially hazardous asteroid,” according to NASA, because my orbital path brings me dangerously close to Earth. In the New Space Age, I will be captured, despun, and towed to a mining depot at a Lagrange equilibrium point, where I will be transformed from a hazard to a resource.



Second Moon

I am Laika, the Earth’s second moon. I am made of compacted space debris, growing bigger and bigger as I clear decommissioned satellites from the orbital environment. Named after the stray dog that was taken from the streets of Moscow and launched into space on Sputnik 2 without any possibility of return, I am made of strays that have been abandoned in orbit.

